Optical Remote Sensing

Time: 90minutes *Total Marks: 40*

Note: Put question number and write the answer
If need, make suitable assumptions and state it clearly when answering

I. *Data Acquisition and Image characteristics* (10 Marks)

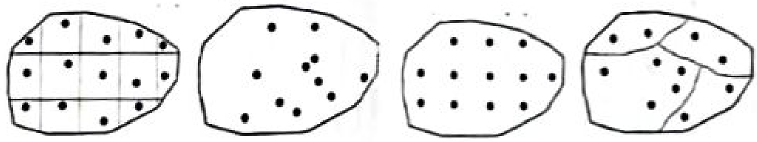
1. Two applications of Landsat SWIR band composites (743/742) 2
2. Brief account (i) Restoring periodic line dropouts, (ii) Least square haze correction 4
3. Difference between linear and histogram stretching 1
4. Complete Surface Reflectance Equation 2
5. ------- filter is used in the pre-processing of satellite imagery to extract (or highlight) road network 0.5
6. Hue is the ------ of the color in a three-dimensional color cube 0.5

II. *Geometric Corrections, Projections and Ground Truth* (13 Marks)

1. List Internal Geometric errors 2
2. List the 4 basic changes observed in pixel (or object) during spatial interpolation 2
3. Give 2 advantages and 2 disadvantages of nearest neighbor pixel interpolation 2
4. Define (i) Datum (ii) Geoid 2
5. (i) What properties are needed to be preserved during projection 1
   (ii) Pair the following 2

Applications	Projections
a. Large area Ocean mapping	e. Zenithal
b. Radio and Seismic mapping	f. Lambert conical
c. Meteorological Charts	g. Lambert Azimuthal Equal Area
d. Aeronautical Charts	h. Equidistant conic

6. Identify the types of ground sampling 2



a b c d

III. *Image Interpretation* and Classification approaches (17 Marks)

1. What prerequisite information is required before performing visual interpretation of satellite data? 3
2. Explain how the elements of visual interpretation can be used to differentiate 4
   (i) Forest and agriculture/plantation (ii) Burnt paddy fields and deep water
   (iii) Evergreen forest and mangrove vegetation (iv) River and Canal
3. Explain why the final number of clusters in ISODATA classification may not be equal to the initial number of clusters. 2
4. Assume a land use land cover type classification has been performed using the Landsat-TM satellite. Subsequently, the classification result is being evaluated by means of a field survey and represented as

	Class types determined from reference source (field data)				
Class types determined from classified map	Classes	Conifer	Hardwood	Water	Totals
	Conifer	50	5	2	57
	Hardwood	14	13	0	27
	Water	3	5	8	16
	Totals	67	23	10	100

Calculate (don't leave answer as fractions)

a. Total accuracy 1

b. Each class producer's and User's accuracy 2

c. Overall classification accuracy 2

5. During a single-band supervised classification, a pixel with a value of 75 is encountered. The user has defined three classes with the following spectral statistics

Class	Min	Max	Mean	Std. Dev.
1	40	90	70	8
2	60	120	85	5
3	10	60	50	6

Briefly explain into which class the pixel will be assigned using: 3

(a) Minimum distance to means classification

(b) Parallelepiped classification

(c) Maximum likelihood classification